

Numerical Analysis

Lab 5

Giuseppe Calì

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Thomas algorithm I

Consider the *tridiagonal* matrix

$$\mathbb{A} = \begin{bmatrix} a_1 & c_1 & & 0 \\ e_2 & a_2 & \ddots & \\ & \ddots & \ddots & c_{n-1} \\ 0 & & e_n & a_n \end{bmatrix}$$

If the $\mathbb{LU}$ decomposition exists, then factors $\mathbb{L}$ and $\mathbb{U}$ are *bidiagonal*, namely

$$\mathbb{L} = \begin{bmatrix} 1 & & & 0 \\ \beta_2 & 1 & & \\ & \ddots & \ddots & \\ 0 & & \beta_n & 1 \end{bmatrix}, \quad \mathbb{U} = \begin{bmatrix} \alpha_1 & c_1 & & 0 \\ & \alpha_2 & \ddots & \\ & & \ddots & c_{n-1} \\ 0 & & & \alpha_n \end{bmatrix}$$

Thomas algorithm II

The unknown coefficients α_i and β_i can be determined by imposing the equality $\mathbb{L}\mathbb{U} = \mathbb{A}$. This yields

$$\alpha_1 = a_1, \quad \beta_i = \frac{e_i}{\alpha_{i-1}}, \quad \alpha_i = a_i - \beta_i c_{i-1}, \quad i = 2, \dots, n.$$

Moreover, due to the bidiagonal structure of $\mathbb{L}$ and $\mathbb{U}$, a special version of the forward and backward substitution algorithms can be applied, so that we obtain

Thomas algorithm

$$(\mathbb{L}\mathbf{y} = \mathbf{b}) \quad y_1 = b_1, \quad y_i = b_i - \beta_i y_{i-1}, \quad i = 2, \dots, n$$

$$(\mathbb{U}\mathbf{x} = \mathbf{y}) \quad x_n = \frac{y_n}{\alpha_n}, \quad x_i = \frac{y_i - c_i x_{i+1}}{\alpha_i}, \quad i = n-1, \dots, 1$$

Exercise 5.1

Consider the tridiagonal matrix $\mathbb{A} \in \mathbb{R}^{10 \times 10}$ defined as

$$\mathbb{A} = \begin{bmatrix} 1 & 11 & & & \\ 102 & 2 & 12 & & \\ & 103 & 3 & 13 & \\ & \dots & \dots & \dots & \\ & & 109 & 9 & 19 \\ & & & 110 & 10 \end{bmatrix}.$$

Then consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ such that $\mathbf{x} = \mathbf{ones}(10, 1)$.

- a Use the Matlab commands `spdiags` and `sparse` to store the matrix in sparse format, and compare the effect of these commands with respect to the commands `diag` and `full`. Read carefully the documentation of the command `spdiags` using the `help` command.
- b Implement the Thomas algorithm and solve the linear system.

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Definition 5.1

Let V be a vector space associated with the field $K = \mathbb{R}$ or $\mathbb{C}$. The map $\|\cdot\|$ from V into $\mathbb{R}$ is a **norm** in V if

- ① $\|\mathbf{v}\| \geq 0 \quad \forall \mathbf{v} \in V$ and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
- ② $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\| \quad \forall \alpha \in K, \forall \mathbf{v} \in V$;
- ③ $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\| \quad \forall \mathbf{v}, \mathbf{w} \in V$.

In $V = \mathbb{R}^n$ we can define the **p -norm**

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \quad 1 \leq p < \infty;$$

when $p = 2$ we recover the **Euclidean norm**. For $p = \infty$ we define the so called **infinity norm**

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

Definition 5.2

Consider $\mathbb{A} \in \mathbb{R}^{m \times n}$. The mapping $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a **matrix norm** if it satisfies the same properties as for vector norms. Moreover, a matrix norm is said **submultiplicative** if it satisfies

$$\textcircled{4} \quad \|\mathbb{A}\mathbb{B}\| \leq \|\mathbb{A}\| \|\mathbb{B}\| \quad \forall \mathbb{A}, \mathbb{B} \in \mathbb{R}^{m \times n}.$$

The matrix 1-norm and matrix ∞ -norm are defined by

$$\|\mathbb{A}\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}| \quad \text{and} \quad \|\mathbb{A}\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$$

namely by the so called **column sum** and **row sum norm**, respectively.

The 2-norm is called **spectral norm** and is defined by $\|\mathbb{A}\|_2 = \sqrt{\lambda_{\max}(\mathbb{A}^T \mathbb{A})}$.

Definition 5.3

The **condition number** with respect to generic norm p is defined as

$$\mathcal{K}_p(A) = \|A\|_p \|A^{-1}\|_p$$

Given a system $A\mathbf{x} = \mathbf{b}$ we consider the problem where the right-hand side term is **perturbed** by vector $\delta\mathbf{b}$

$$A(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$$

$\mathcal{K}_p(A)$ is a **measure of the sensitivity** of the solution $\mathbf{x}$ to changes in $\mathbf{b}$ and it holds

$$\frac{\|\delta\mathbf{x}\|_p}{\|\mathbf{x}\|_p} \leq \mathcal{K}_p(A) \frac{\|\delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}$$

- $\mathcal{K}_p(A) = 10^k$: you **may lose** up to k digits of accuracy in the solution
- MATLAB: `cond(A,p)` (`cond(A)`, $p = 2$ by default)

Exercise 5.2

Consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \varepsilon \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

where $0 < \varepsilon \ll 1$.

- a Compute $\mathcal{K}_p(\mathbb{A})$ for $p = 1, 2, \infty$ without using MATLAB. Check the results in MATLAB for some values of ε ($1e-6, 1e-8$).
- b For $\varepsilon = 1e-6$, and $\alpha = 10^{-12}, 10^{-6}$, consider the perturbation term $\delta\mathbf{b} = [0 \ 0 \ \alpha]^T$, and compute the associated perturbation on the solution $\mathbf{x}$.
- c Consider now $\delta\mathbf{b} = [\alpha \ 0 \ 0]^T$ and compute the associated perturbation on the solution $\mathbf{x}$

Exercise 5.3

Consider the Hilbert matrix $\mathbb{H}_n$ of order n

$$\mathbb{H}_n = \begin{bmatrix} 1 & 1/2 & \cdots & 1/n \\ 1/2 & 1/3 & \cdots & 1/(n+1) \\ \vdots & \vdots & \ddots & \vdots \\ 1/n & 1/(n+1) & \cdots & 1/(2n-1) \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad n = 10, 20, 40.$$

- a Estimate $\mathcal{K}_2(\mathbb{H}_n)$ using the technique of perturbing a linear system with $\mathbb{H}_n$ as coefficient matrix. Consider the right hand side term $\mathbf{b}$ such that $\mathbf{x} = \text{ones}(n, 1)$.
- b Compute $\mathcal{K}_2(\mathbb{H}_n)$ using MATLAB command `cond`.
- c What can you say about the **accuracy of the solution** of the generic linear system $\mathbb{H}_n \mathbf{x} = \mathbf{b}$? How large can you take n before the relative error is 1 (i.e. there are no significant digits in the solutions)?

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Iterative schemes

According to a generic iterative **method**, the solution $\mathbf{x}$ to the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ is approximated by

$$\mathbf{x}^{(k+1)} = \mathbb{B}\mathbf{x}^{(k)} + \mathbf{g}, \quad \mathbb{B} : \text{iteration matrix}$$

such that

$$\mathbf{x} = \mathbb{B}\mathbf{x} + \mathbf{g} \quad \text{and} \quad \mathbf{x} = \lim_{k \rightarrow \infty} \mathbf{x}^{(k)}$$

(consistency)

(convergence)

Usually $\mathbf{x}$ is not reached in a finite number of iterations, so a **stopping criterion** is applied

$$\bullet \text{ increment } \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| \leq \varepsilon \qquad \bullet \text{ residual } \|\mathbf{b} - \mathbb{A}\mathbf{x}^{(k)}\| \leq \varepsilon.$$

If the method is consistent

$$\mathbf{x}^{(k+1)} = \mathbb{B}\mathbf{x}^{(k)} + \mathbf{g} \text{ converges} \iff \rho(\mathbb{B}) < 1$$

$R(\mathbb{B}) = -\log \rho(\mathbb{B})$ **asymptotic convergence rate**.

Jacobi and Gauss-Seidel methods I

The square matrix $\mathbb{A}$ can be written as $\mathbb{A} = -\mathbb{E} + \mathbb{D} + (\mathbb{A} - \mathbb{D} + \mathbb{E})$ where

- $-\mathbb{E}$: lower triangular part starting from the first subdiagonal (MATLAB: $-E = \text{tril}(A, -1)$)
- $-\mathbb{F}$: upper triangular part starting from the first superdiagonal (MATLAB: $-F = \text{triu}(A, 1)$)
- $\mathbb{D}$: diagonal matrix extracted from $\mathbb{A}$ (MATLAB: $\mathbb{D} = \text{diag}(\mathbb{A})$)

Jacobi method

$$\mathbb{B}_J = \mathbb{I} - (\mathbb{D})^{-1}\mathbb{A}$$
$$\mathbf{g}_J = \mathbb{D}^{-1}\mathbf{b}$$

Gauss-Seidel method

$$\mathbb{B}_{GS} = (\mathbb{D} - \mathbb{E})^{-1}\mathbb{F}$$
$$\mathbf{g}_{GS} = (\mathbb{D} - \mathbb{E})^{-1}\mathbf{b}$$

Convergence Jacobi

$\mathbb{A}$ is strictly diagonally
dominant by rows or by
columns

$\implies$

Jacobi and Gauss-Seidel
methods converge for any initial
guess $\mathbf{x}^{(0)}$

Convergence Gauss-Seidel

$\mathbb{A}$ is symmetric
and positive
definite

$\implies$

Gauss-Seidel method converges
for any initial guess $\mathbf{x}^{(0)}$

Exercise 5.4

Consider the tridiagonal matrix $\mathbb{A} \in \mathbb{R}^{10 \times 10}$ defined as

$$\mathbb{A} = \begin{bmatrix} 3 & -2 & & & \\ -1 & 3 & -2 & & \\ & \dots & \dots & \dots & \\ & & -1 & 3 & -2 \\ & & & -1 & 3 \end{bmatrix}.$$

Consider also the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ such that $\mathbf{x} = \mathbf{ones}(10, 1)$.

- a Implement the Jacobi and Gauss-Seidel methods and solve the linear system.
- b Do both the Jacobi and the Gauss-Seidel methods converge? Numerically confirm the relation between their spectral radii.
- c Apply both methods, with $\mathbf{x}^{(0)} = \mathbf{0}$ and with a tolerance of 10^{-12} , and compare the required number of iterations.

Exercise 5.5 (Exam Sep 7, 2023)

Consider the matrices $\mathbb{A}$ and $\mathbb{B}$ defined by the following MATLAB commands:

```
n=100;  
A=-19*diag(ones(n-2,1),-2) - 7/3*diag(ones(n,1)) +  
+3*diag(ones(n-1,1),1) + 3*diag(ones(n-1,1),-1)+  
+57/3*diag(ones(n-2,1),-2);  
C=-7*diag(ones(n,1)) + 2*diag(ones(n-2,1),-2)  
+2*diag(ones(n-2,1),2);  
B=C'*C;
```

- a Solve Numerically the linear systems $\mathbb{A}\mathbf{x} = \mathbf{a}$, $\mathbb{B}\mathbf{x} = \mathbf{b}$ with $\mathbf{a} = A * \text{ones}(n, 1)$, $\mathbf{b} = B * \text{ones}(n, 1)$. Resort to the cheapest direct method in terms of computational cost.
- b Compute the error associated with the two approximations computed at the previous point, in terms of the infinity norm.

Exercise 5.6 (Exam June 17, 2024)

Consider the family of matrices $\mathbb{A}_n(a) \in \mathbb{R}^{n \times n}$ defined by:

$$\mathbb{A}_n(a) = \begin{bmatrix} 1 & -11 & 0 & 0 & \dots \\ -11 & 1 & -12 & 0 & \dots \\ 0 & -12 & 1 & -13 & \dots \\ 0 & 0 & -13 & \ddots & \ddots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix} + a\mathbb{I}_n.$$

where $\mathbb{I}_n$ is the identity matrix of order n .

- a Resorting to the MATLAB commands `diag` and `eye`, store the matrix $\mathbb{A}_7(32)$. Report the commands used.
- b Resorting to the MATLAB command `lu`, compute the LU factorization of $\mathbb{A}_7(32)$ and answer to the following questions (use the commands `nnz` and `spy`), providing justifications:
 - 1 Was the pivoting performed?
 - 2 did fill-in occur?

Include the MATLAB commands used in your answer.

- c Set $\mathbf{x}_{\text{ex}} = [1, 1, 1, \dots]^T \in \mathbb{R}^n$ and consider the linear system $\mathbb{A}_n(a)\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = \mathbb{A}_n(a)\mathbf{x}_{\text{ex}}$. Use the provided `jacobi.m` function to solve the linear system for $n = 7$ and $a = 1$ and compute the error $\eta_7(1) = \|\mathbf{x}_{\text{ex}} - \mathbf{x}\|_2$. Set `tol=1e-6` and `nmax=100`. Choose the initial guess $\mathbf{x}^{(0)} = \mathbf{0}$. Comments the results and report the MATLAB commands.
- d Use the provided `jacobi.m` function to solve the linear system for $n = 7$ and $a = 32$ and compute the error $\eta_7(32) = \|\mathbf{x}_{\text{ex}} - \mathbf{x}\|_2$. Set `tol=1e-6` and `nmax=100`. Choose the initial guess $\mathbf{x}^{(0)} = \mathbf{0}$. Compare the results with the previous case and discuss them considering the theoretical expectations.

Exercise 5.7 (Exam July 21, 2021)

Consider the matrix $\mathbb{A}$:

$$\mathbb{A}_n(a) = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & 1 \\ 0 & -1 & 4 \end{bmatrix} + a\mathbb{I}_n.$$

Set $\mathbf{b}$ such that $\mathbf{x}_{\text{ex}} = [1/2, 1/3, 1/4]^T$. With the aim of solving the linear system $\mathbb{A}_n(a)\mathbf{x} = \mathbf{b}$, with an iterative method, set $\mathbf{x} = [0, 0, 0]^T$ and $N_{\text{max}} = 100$.

Exam July 21, 2021

- a Using the `jacobi.m` function solve the linear system with the Jacobi iterative method with the following tolerances: $tol = 10^{-2}, 10^{-3}, 10^{-4}$. Report the obtained solutions and the number of iterations required for each case and the MATLAB instructions.
- b Comment the results obtained in the previous point.
- c By using the function `gs.m` solve the linear system with the Gauss-Seidel iterative method with tolerance $tol = 10^{-4}$. Report the obtained solution and the number of iterations required. Include the MATLAB instructions.
- d Regarding point a with $tol = 10^{-4}$ and point c, explain why the two methods converge and compare the convergence of the two methods, justifying their behavior in light of the theory.
- e Referring to the iteration matrices constructed in `jacobi.m` and `gs.m`, calculate the spectral radius of the two methods for the proposed exercise and compare the results in light of the findings obtained in point d.

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4 Homework

Homework 5.1

Consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 10^{10} & 1 & 1 \\ 10^{10} & 1 & 1 & 10^{10} \\ 1 & 1 & 10^{-10} & 1 \\ 1 & 10^{10} & 1 & 10^{10} \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 10^{10} + 3 \\ 2 \cdot 10^{10} + 2 \\ 3 + 10^{-10} \\ 2 \cdot 10^{10} + 2 \end{bmatrix}$$

such that the exact solution is $\mathbf{x} = [1 \ 1 \ 1 \ 1]^T$.

- a Solve the system using LU decomposition with pivoting
- b Compute the infinity norm of the error associated with the solution of the previous item
- c Can we apply the Cholesky decomposition to matrix $\mathbb{A}$?
- d Is the normalized residual $\|\mathbf{b} - \mathbb{A}\mathbf{x}\|/\|\mathbf{b}\|$ a good estimator for the relative error? Motivate your answer.

Homework 5.2

Compute the condition number and the determinant of the following matrices

```
» A = diag(0.1*ones(15,1));
```

```
» B = triu(rand(15),14)*1e5 + diag(0.1*ones(15,1));
```

Is, in general, the determinant a good measure of the condition number of the matrix?

Homework 5.3

Consider the following matrix

$$\mathbb{A} = \begin{bmatrix} 1.001 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1.001 & 1 & 1 \\ 1 & 1.001 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1.001 \end{bmatrix}.$$

- a Compute the condition number $\mathcal{K}_{\infty}(\mathbb{A})$ using MATLAB command `cond`.
- b Then compute $\mathcal{K}_{\infty}(\mathbb{A})$ using its definition.
- c Finally estimate $\mathcal{K}_{\infty}(\mathbb{A})$ by **perturbing a suitable linear system** having $\mathbb{A}$ as coefficient matrix.

Homework 5.4

Consider the linear systems $\mathbb{A}_i \mathbf{x} = \mathbf{b}_i$, $i = 1, \dots, 4$ with

$$\begin{aligned}\mathbb{A}_1 &= \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} & \mathbb{A}_2 &= \begin{bmatrix} 2 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \\ \mathbb{A}_3 &= \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} & \mathbb{A}_4 &= \begin{bmatrix} 1 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}\end{aligned}$$

with $\mathbf{x} = [1 \ 2 \ 3]^T$ and $\mathbf{b}_i = \mathbb{A}_i \mathbf{x}$.

- a Write a MATLAB function that implements the generic iterative scheme $\mathbf{x}^{(k+1)} = \mathbb{B} \mathbf{x}^{(k)} + \mathbf{g}$. Use the stopping criterion based on the increment
- b Study the convergence for Jacobi and Gauss-Seidel methods. Check the results using the function at the previous item, setting $\mathbf{x}_0 = [0, 0, 0]^T$, $\text{tol} = 1.e-6$, $\text{maxit} = 100$

Homework 5.5

Let the 10×10 square matrix $\mathbb{A}$ be

$$\mathbb{A} = \begin{bmatrix} 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 \end{bmatrix}$$

Consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{x} = [1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]^T$.

- 1 Study the convergence of the Jacobi and Gauss-Seidel methods by investigating the relation between spectral radius of the two iteration matrices
- 2 Check the results using the functions implemented in 5.3 and choosing $\mathbf{x}_0 = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T$, $\text{tol} = 1.e-12$, $\text{maxit} = 500$

Homework 5.6

Consider the linear system $\mathbb{A}_3 \mathbf{x} = \mathbf{b}$ with

$$\mathbb{A}_3 = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b}_3 = \mathbb{A}_3 \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

- a Apply Jacobi method to compute the solution with a tolerance of 10^{-5} and 10^{-8} . What do you observe?
- b Compute the iteration matrix $\mathbb{B}_J$. Evaluate its spectral radius $\rho(\mathbb{B}_J)$ and $(\mathbb{B}_J)^3$. Relying on the results, motivate what you found at the previous point.

Homework 5.7

Consider a linear system whose coefficient matrix is obtained with the commands

```
» B = rand(5) + diag(10*ones(5,1));
```

```
» A = B*B';
```

Analyze the convergence of Jacobi and Gauss-Seidel methods.

Homework 5.8

Consider a system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 3 & -1 & 0 & 0 & 0 & -1 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & 0 \\ 0 & -1 & 0 & -1 & 3 & -1 \\ -1 & 0 & 0 & 0 & -1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

- a Study the convergence of Jacobi and Gauss-Seidel methods if applied to the system $\mathbb{A}\mathbf{x} = \mathbf{b}$.
- b Compute the spectral radii of the iteration matrices of both methods.
- c How many iterations are needed to obtain the solution with a tolerance of 10^{-8} starting from $\mathbf{x}^{(0)} = \mathbf{0}$?

Homework 5.9

Consider a system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 50 & 1 & 2 \\ 1 & 5 & 2 \\ 2 & 2 & 7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

- a Compute $\|\mathbb{A}\|_{\infty}$ and $\|\mathbb{A}\|_1$.
- b Is $\mathbb{A}$ symmetric and positive definite?
- c Compute the LU decomposition using `lu_decomposition`.
- d Compute $\mathcal{K}_2(\mathbb{A})$. Is $\mathbb{A}$ well-conditioned?
- e Which methods (direct and iterative) can be employed for solving the system? Motivate. If Jacobi method is in that list, estimate both the convergence rate and the error after 4 iterations starting with $\mathbf{x}^{(0)} = \mathbf{0}$. Numerically validate the results.

Homework 5.10 (Exam February 14, 2012)

Let $\mathbb{A} \in \mathbb{R}^{n \times n}$, $n > 3$ be defined as

$$\mathbb{A} = \begin{bmatrix} 1 & 0 & \cdots & 0 & (1/\alpha) \\ 0 & 1 & & & 0 \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}, \quad 0 < \alpha < 1.$$

- a Write the iteration matrix $\mathbb{J}$ for the Jacobi method applied to the solution of the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ and compute $\mathbb{J}^2$ and $\mathbb{J}^3$.
- b Let $\alpha = 10^{-4}$, $\mathbf{x}^{(0)} = \mathbf{b} = [1, 1, \dots, 1]^T$. Will the Jacobi method converge? How many iterations will be required to ensure that the infinity norm of the error is less than 10^{-9} ?
- c Let $\mathbb{B} = \mathbb{A}^T \mathbb{A}$ and consider the linear system $\mathbb{B}\mathbf{x} = \mathbf{b}$. Compute the iteration matrix $\mathbb{G}$ for the Gauss-Seidel method. Will the Gauss-Seidel method converge?
- d Repeat point b applying Gauss-Seidel method.

References I